



Data Warehouse Project Report

Analyzing Customer Reviews of Bank Agencies in Morocco using a Modern Data Stack

Master in Information Systems and Intelligent Systems

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Abstract

This project explores the application of a modern data warehouse stack to analyze customer reviews of bank agencies in Morocco. By leveraging tools such as Python, Airflow, DBT, and Looker Studio, we develop an automated pipeline that transforms unstructured Google Maps reviews into actionable insights. The system performs sentiment analysis and topic modeling to evaluate customer satisfaction across different bank branches. This work demonstrates how data engineering and business intelligence can be combined to improve customer experience in the banking sector, while serving as a practical implementation of concepts studied in the Master's program in Information Systems and Intelligent Systems.

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1 Motivation

In a context where customer experience becomes a decisive factor for the competitiveness of companies, banks must pay close attention to their customers' feedback. Reviews published on Google Maps constitute a valuable source of information, although often underutilized due to their unstructured nature.

This project is part of my Master's training in Information Systems and Intelligent Systems and aims to demonstrate how an approach based on a modern Data Warehouse can transform this raw data into actionable insights. By leveraging current tools such as Python, Airflow, DBT, and Looker Studio, the goal is to create an automated pipeline capable of analyzing customer sentiment, identifying strengths and weaknesses of bank branches, and providing a clear and dynamic view of customer satisfaction.

This work represents an opportunity to combine my skills in data engineering and intelligent analysis to address a real need in the banking sector.

2 Introduction

In the digital age, customer reviews on platforms such as Google Maps have become critical sources of information for businesses aiming to improve their services. These reviews offer unfiltered and direct feedback from clients, reflecting their real-life experiences with products, services, and support. In the context of the banking sector in Morocco, thousands of reviews are posted across different branches, yet remain largely underexploited due to their unstructured nature.

This project aims to harness the power of these customer reviews by building a modern data pipeline that automates the collection, transformation, analysis, and visualization of the data. By applying natural language processing techniques such as sentiment analysis and topic modeling, the project seeks to extract key insights related to customer satisfaction, common complaints, and areas of excellence.

The analysis focuses on banks operating in Morocco and their respective agencies. The end goal is to evaluate branch performance based on user sentiment, identify recurring issues, and provide actionable recommendations to enhance customer experience. This initiative reflects the growing importance of data-driven decision-making in customer relationship management.

Through this project, I will demonstrate how a modern data stack — composed of tools such as Python, Apache Airflow, DBT, PostgreSQL, and Looker Studio — can be used to deliver a robust and automated solution for extracting value from unstructured textual data. This work not only contributes to my academic development in the Master's program in Information Systems and Intelligent Systems but also addresses real-world challenges in the Moroccan banking industry.

3 Chapter 1 – Data Warehouse: Concepts and Architecture

3.1 What is a Data Warehouse?

A **Data Warehouse** is a centralized computer system designed to collect, store, consolidate, and analyze data coming from heterogeneous and multiple sources. This data is generally extracted from different operational systems, then cleaned, transformed, and organized according to a specific structure using ETL (Extract, Transform, Load) processes.

The primary goal of a Data Warehouse is to provide a single, consistent database that allows decision-makers easy access to historical, reliable, and structured data for performing decision analysis, strategic reporting, and dashboards.

Unlike transactional databases, a Data Warehouse is optimized for complex queries and multidimensional analysis, thus facilitating the exploration of trends, understanding customer behavior, operational performance, and making informed strategic decisions.

3.2 Key Advantages

- **Data Consistency:** The Data Warehouse enables unification of formats, structures, and business rules of data coming from various sources. This harmonization guarantees better data quality, avoids inconsistencies, and facilitates reliable comparison and analysis across different departments and systems.
- **Improved Decision Making:** By centralizing historical and current data in a single platform, the Data Warehouse offers a comprehensive and consistent view of the company's activities. This consolidated visibility allows decision-makers to quickly access relevant information, identify trends, and make informed decisions based on precise and complete analyses.
- **Cost Optimization:** Automation of extraction, transformation, and loading (ETL) processes, along with data centralization, reduces duplication of effort and human errors. Additionally, the Data Warehouse facilitates automatic generation of reports and metrics, lowering costs related to manual processing, maintaining disparate systems, and accelerating the decision-making cycle.

The global Data Warehouse market continues to evolve rapidly:

- In 2018, the global Data Warehouse market was estimated at approximately 13 billion US dollars, reflecting growing adoption by companies aiming to improve their data management and analysis.

- The sector is experiencing rapid growth, with a projected Compound Annual Growth Rate (CAGR) of 12% through 2025, highlighting the increasing importance of analytical solutions and data infrastructures in modern businesses.
- In 2023, nearly 47% of IT leaders reported having migrated all or part of their Data Warehouse to cloud solutions, reflecting a strong move toward flexibility, scalability, and reduced costs associated with on-premises infrastructures.
- Among major players, Snowflake holds a significant position in the global Data Warehouse market, with about 19% market share, thanks to its cloud-native architecture and advanced data analytics and management capabilities.

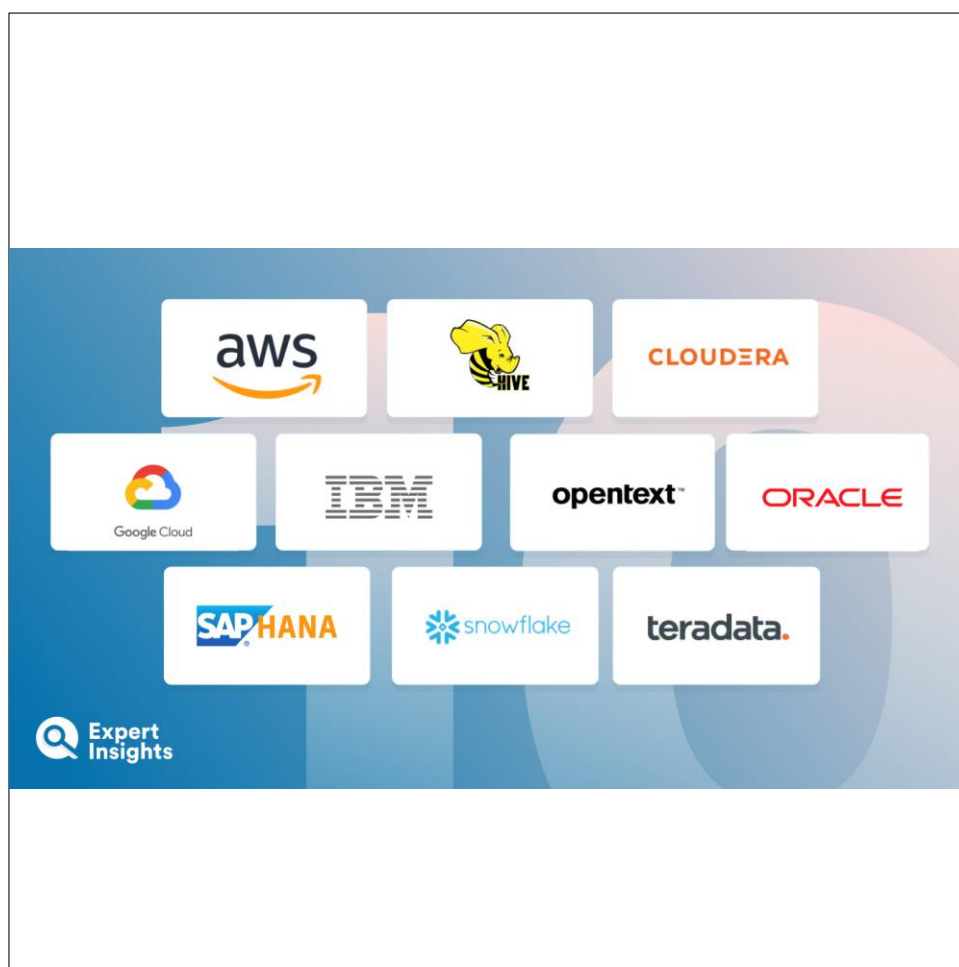


Figure 1: Major Data Warehouse Vendors (2023)

3.3 Typical Architecture of a Data Warehouse

A classic architecture includes three layers:

1. **Data Sources:** These are the various origins of data, such as transactional databases, external APIs, or flat files. These sources provide raw data that will later be integrated into the Data Warehouse.
2. **Processing Zone (ETL):** This step corresponds to the extraction, transformation, and loading (ETL) processes. Data is cleaned, enriched, and formatted to ensure its quality and consistency before storage.
3. **Storage Zone:** Transformed data is stored in a structure optimized for analysis, often in the form of a star schema or snowflake schema, facilitating fast and efficient analytical queries.
4. **BI / Reporting Tools:** This layer groups the Business Intelligence tools used to visualize, analyze, and generate reports from stored data. It includes interactive dashboards and analytical queries used by decision-makers.

3.4 Star Schema

The star schema is composed of:

- **A fact table:** This central table contains the main quantitative measures such as the number of reviews, the score given by customers, or the date of the comment. It represents the events to analyze.
- **Dimension tables:** These tables provide contextual information allowing analysis of facts from different perspectives, for example, the concerned bank branches, their geographic location, the type of sentiment expressed in reviews, etc. They facilitate queries and aggregations across multiple criteria.

3.5 1.6 Conclusion

This chapter highlights the strategic importance of the Data Warehouse in analyzing complex data such as customer reviews. It sets the stage for the modeling and implementation of our own analytical architecture for bank branches in Morocco.

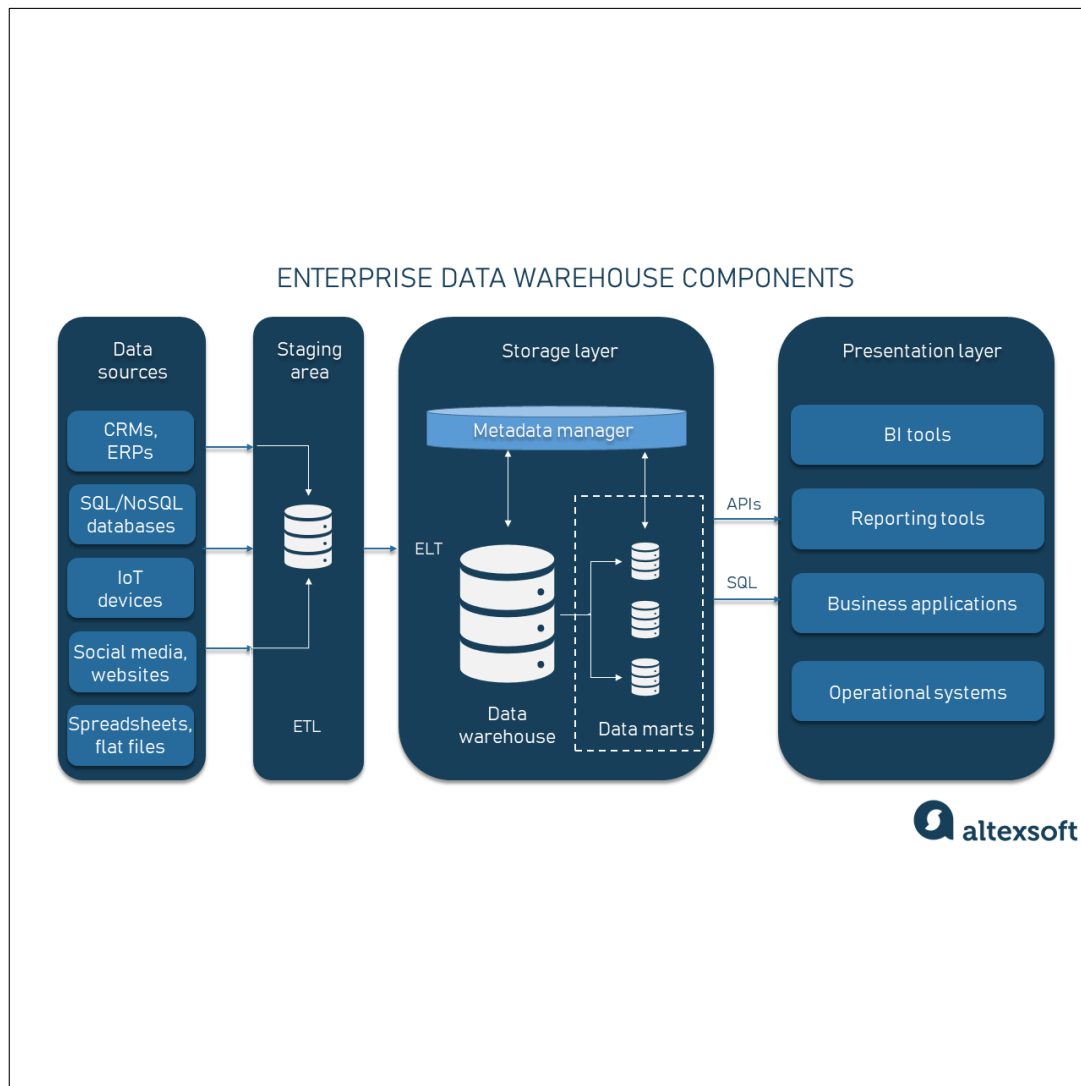


Figure 2: General Architecture of a Data Warehouse

4 Chapter 2 – Project Overview

4.1 Title and Objective

The project, titled **”Analyzing Customer Reviews of Bank Agencies in Morocco using a Modern Data Stack”**, focuses on harnessing the wealth of information embedded within online customer feedback to improve banking services. Specifically, it targets the collection, processing, and analysis of customer reviews posted on Google Maps for various bank branches distributed throughout Morocco.

The primary objective is to transform these raw, unstructured textual data into actionable insights that can guide strategic decision-making for bank management. By applying advanced data processing techniques such as natural language processing (NLP), sentiment analysis, and topic modeling, the project seeks to uncover prevailing themes, gauge customer sentiment, and identify strengths and weaknesses of each branch.

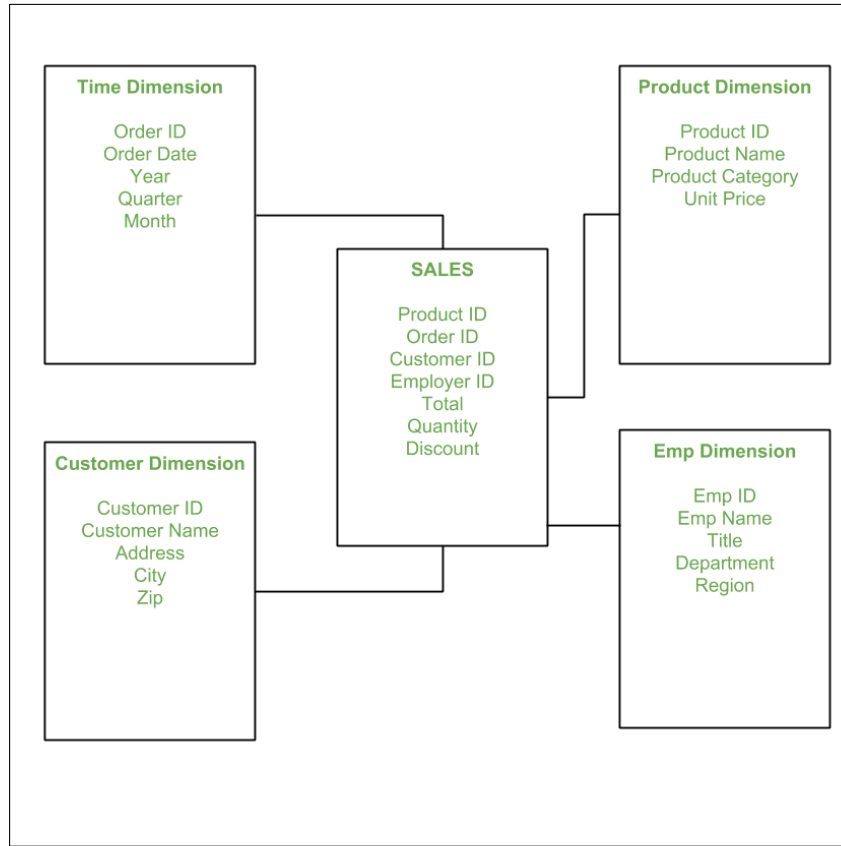


Figure 3: Example of a Star Schema

Moreover, the project aims to establish a robust, automated data pipeline using a modern data stack comprising tools like Python for data processing, Apache Airflow for workflow orchestration, DBT (Data Build Tool) for data transformation, and Looker Studio for dynamic data visualization and reporting.

This approach intends to provide bank executives and operational teams with a clear, data-driven understanding of customer experience, enabling targeted improvements in service quality, operational efficiency, and customer satisfaction across the network of bank agencies in Morocco.

4.2 Project Scope

In today's competitive banking environment, customer feedback plays a critical role in shaping service delivery and improving operational excellence. Banks across Morocco receive thousands of reviews daily on Google Maps, a platform where customers freely express their opinions about branch services, staff behavior, wait times, facilities, and overall satisfaction. These reviews constitute a vast repository of unstructured data that, if properly harnessed, could provide invaluable insights to decision-makers aiming to enhance customer experience and branch performance.

However, this data is inherently scattered and unstructured. Reviews come in vary-

ing formats, languages, lengths, and tones, making manual analysis infeasible. Important trends and customer sentiments often remain hidden within the noise, delaying or impeding proactive service improvements. This project aims to bridge this gap by centralizing these reviews into a unified Data Warehouse, cleaning and enriching the raw textual data, and applying advanced analytical techniques to extract meaningful information.

Specifically, the scope encompasses the following key objectives:

- **Sentiment Analysis:** Utilize natural language processing (NLP) models to classify reviews based on customer sentiment—positive, neutral, or negative. Tracking sentiment trends over time will reveal shifts in customer satisfaction and help identify emerging issues before they escalate.
- **Topic Modeling:** Apply unsupervised machine learning algorithms such as Latent Dirichlet Allocation (LDA) to detect recurrent themes or topics within reviews. This enables banks to pinpoint specific areas frequently praised or criticized, such as staff professionalism, queue management, ATM availability, or digital services.
- **Branch Performance Evaluation:** Develop a quantitative scoring mechanism to rank bank branches based on aggregated customer sentiment and thematic analysis. This provides a transparent and data-driven benchmark, facilitating internal comparisons and resource allocation.
- **Customer Experience Metrics:** Extract and monitor key drivers of customer satisfaction and dissatisfaction by correlating sentiment with thematic categories and metadata such as review date, branch location, and service type. This multi-dimensional insight supports targeted improvement strategies.

By accomplishing these objectives, the project not only leverages publicly available data for strategic advantage but also establishes a scalable framework for ongoing customer experience monitoring. Ultimately, it empowers Moroccan banks to become more responsive, customer-centric, and competitive.

4.3 Technology Stack

The successful realization of the project hinges on adopting a modern, scalable, and flexible technology stack. Each component is selected to address specific stages of the data lifecycle, from collection to presentation, ensuring seamless integration and efficient automation.

- **Data Collection:** Python is the primary language used for data extraction due to its rich ecosystem of libraries for web scraping and API interaction. The Google Maps API facilitates direct access to review data where possible, providing structured endpoints for reliable extraction. In cases where API limitations apply, web scraping frameworks such as BeautifulSoup or Scrapy are employed to crawl and parse reviews from Google Maps web pages, ensuring comprehensive data coverage.
- **Orchestration:** Apache Airflow serves as the workflow orchestrator, enabling the automation, scheduling, and monitoring of complex data pipelines. Airflow's Directed Acyclic Graph (DAG) architecture allows for modular pipeline design, retry mechanisms, and alerting systems to ensure data freshness and operational resilience.
- **Data Storage:** PostgreSQL is chosen as the relational database management system for the Data Warehouse. Its robustness, advanced indexing, and support for JSON data types make it well-suited for storing both structured and semi-structured data. The warehouse schema is designed to optimize query performance for analytical workloads.
- **Data Transformation:** DBT (Data Build Tool) facilitates the transformation and modeling of raw data within the warehouse. DBT enables version-controlled SQL-based transformations, promoting transparency, reproducibility, and easy maintenance of complex data logic.
- **Analysis and Reporting:** Looker Studio (formerly Google Data Studio) is employed for creating interactive dashboards and reports. Its connectivity with PostgreSQL and ease of use allow for rapid visualization development, empowering stakeholders to explore insights dynamically.
- **Version Control:** GitHub manages source code, configuration files, and documentation, enabling collaboration, change tracking, and integration with continuous integration/continuous deployment (CI/CD) workflows.

The synergy of these tools provides an end-to-end solution that is scalable, maintainable, and adaptable to evolving business needs and data volumes.

4.4 Project Roadmap

The project follows a structured roadmap divided into six main phases, each building on the outputs of the previous stage to ensure progressive refinement and validation of the solution.

1. **Data Collection:** The initial phase involves developing scripts and workflows to extract customer reviews directly from Google Maps using web scraping techniques. We utilize Selenium to render dynamic content and BeautifulSoup to parse and extract structured data such as review date, rating, reviewer identity (if publicly available), and the review text. This approach allows us to bypass limitations related to data accessibility while ensuring completeness and accuracy. All scraping activities are performed in accordance with ethical guidelines and legal boundaries. The collected raw data is ingested into a PostgreSQL database hosted on a cloud server via Aiven, providing secure, scalable, and reliable storage while preserving the original data formats for maximum fidelity.



2. **Data Cleaning and Transformation:** The raw dataset is then subjected to rigorous cleaning processes. Duplicate reviews are identified and removed to avoid skewing analyses. Text normalization includes converting to lowercase, removing punctuation, stopwords, and applying lemmatization or stemming to reduce lexical variability. Missing or inconsistent metadata are handled through imputation or exclusion based on relevance. The enriched dataset undergoes sentiment classification using pretrained NLP models fine-tuned on banking-related customer feedback. Topic modeling via LDA or similar algorithms segments reviews into thematic clusters, revealing common subjects discussed by customers.



3. **Data Modeling:** A well-designed data model is essential for analytical efficiency. A star schema is constructed, centered on a fact table (fact reviews) that records each review as an event with measurable attributes such as sentiment score, review length, and date. Surrounding this are dimension tables including dim bank (bank identity), dim branch (branch-specific details), dim location (geographical data), and dim sentiment (sentiment categories). This decision-oriented star schema, representing the cleaned and transformed data optimized for analysis, is deployed within a PostgreSQL database hosted on Aiven, enabling robust, cloud-based access for analytics. DBT automates incremental loading and transformation, ensuring that analytical tables remain up to date with minimal manual intervention.



Data Analysis and Visualization: This phase translates data into actionable insights. An OLAP (Online Analytical Processing) cube structure is designed to enable multidimensional analysis of the review data. The data model includes dimension tables such as `dim_bank`, `dim_branch`, `dim_location`, and `dim_sentiment`, and a centralized `fact_reviews` table containing metrics like review date, sentiment, topic analysis, and ratings. This schema supports flexible slicing and dicing across key attributes such as branch, location, sentiment category, and time.

Interactive dashboards are then developed in Looker Studio, directly leveraging these OLAP cubes to present key performance indicators—such as overall sentiment trends, volume of reviews per branch, common complaint topics, and branch rankings based on aggregated sentiment scores.



The architecture of project

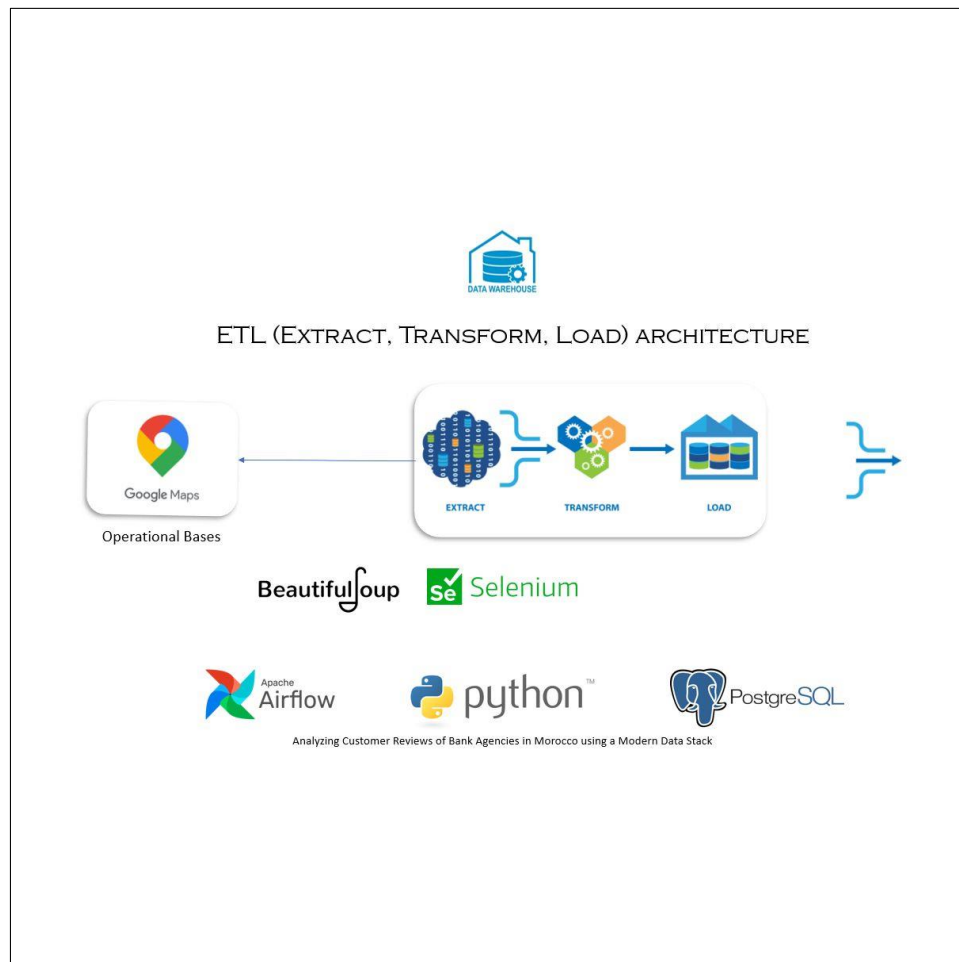
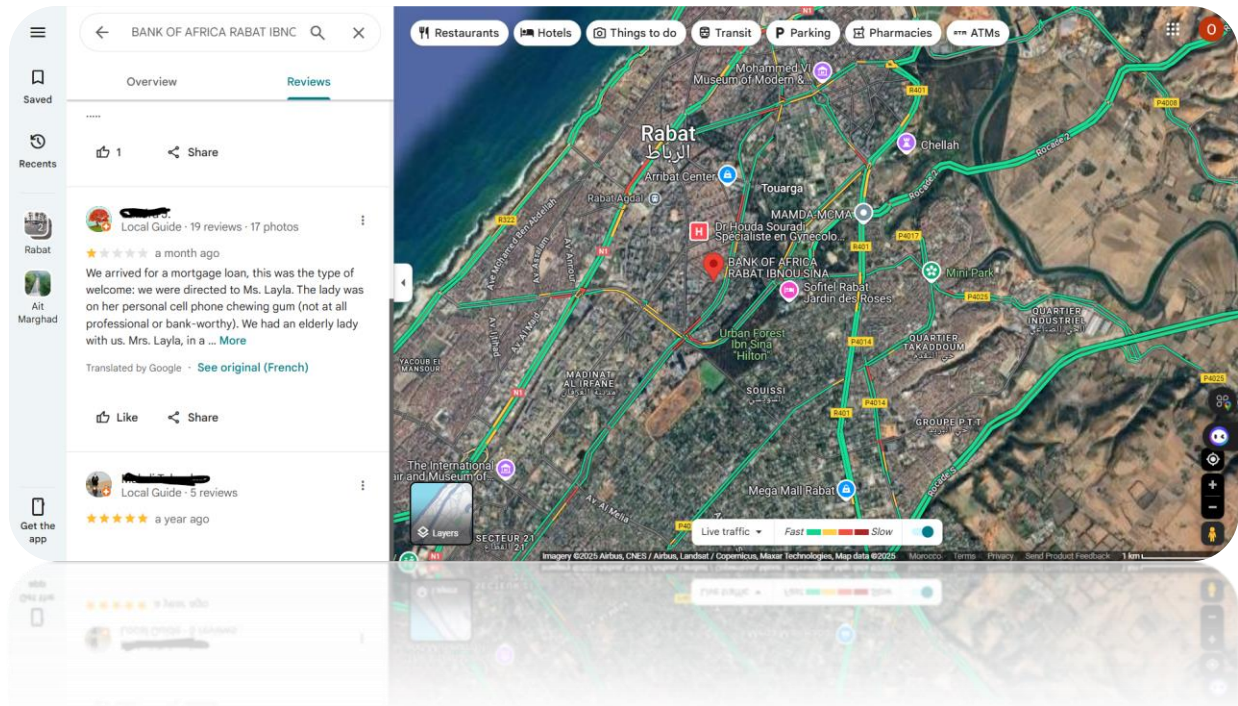


Figure 4: Overview of the Data Collection Workflow: API requests and web scraping to collect customer reviews and metadata, followed by storage in the cloud-hosted PostgreSQL database.

Phase 1: Data Collection (Scraping Google Maps Reviews)



Pourquoi Google Maps ?

Google Maps est bien plus qu'un simple outil de navigation. C'est une plateforme riche en contenu généré par les utilisateurs, notamment à travers son système d'évaluation et de commentaires. Chaque établissement référencé, y compris les agences bancaires, peut recevoir des avis de clients qui partagent leur expérience.

Ces avis contiennent généralement :

- Une note (de 1 à 5 étoiles)
- Un commentaire textuel
- La date de publication de l'avis
- Des informations sur l'établissement concerné (nom, adresse, localisation géographique)

Dans le cadre de ce projet, qui vise à analyser les avis clients des agences bancaires au Maroc, Google Maps représente une **source de données pertinente et abondante**. On y retrouve :

- Une large couverture géographique des agences bancaires marocaines
- Des retours d'expérience réels, utiles pour l'analyse de la satisfaction client
- Une combinaison de données structurées (notes, dates) et non structurées (commentaires)

L'objectif est donc d'exploiter ces avis à l'aide de techniques de web scraping, notamment avec Selenium et BeautifulSoup, afin de collecter automatiquement les données nécessaires pour l'analyse. Cette extraction constitue la première phase du processus ETL (Extract, Transform, Load).

```

def extract_data(huargs):
    ti = huargs['ti']
    bank_urls_dict = ti.xcom_pull(task_id='extract_links', key='all_links')
    driver = initialize_driver()
    all_data = {}

    for search_url, bank_urls in bank_urls_dict.items():
        print(f"Processing URLs from: {search_url}")

        # Set correct bank name based on search url
        if "banque" in search_url:
            bank_name = "Banque"
        elif "banque" in search_url:
            bank_name = "Banque Populaire"
        else:
            bank_name = "Unknown Bank"

        for index, url in enumerate(bank_urls, start=1):
            print(f"Scraping de l'URL (index): {url}")
            driver.get(url)
            time.sleep(5)

            # Extract bank address
            try:
                address_element = WebDriverWait(driver, 10).until(
                    EC.presence_of_element_located((By.CSS_SELECTOR, "div.IndVte.fontbod"))
                )
                address = address_element.text.strip()
            except TimeoutException:
                address = "Non trouvée"

            # Click on "Avis" button to reveal reviews
            try:
                buttons = WebDriverWait(driver, 10).until(
                    EC.presence_of_all_elements_located((By.CSS_SELECTOR, "div.RdPd bu"))
                )
                buttons[1].click()
                time.sleep(3)
            except (TimeoutException, IndexError):
                print(f"⚠️ 'Avis' button not found for {bank_name}")

            # Extract reviews
            def extract_reviews():
                reviews = driver.find_elements(By.CSS_SELECTOR, "div.RdPd span.w17pd")
                return [review.text for review in reviews]

            all_reviews = set()
            scroll_count = 0
            max_scroll = 2
            retry_count = 0

            while scroll_count < max_scroll:
                current_reviews = extract_reviews()
                unique_reviews = set(current_reviews) - all_reviews
                all_reviews.update(unique_reviews)

                print(f"Scroll {scroll_count + 1}: Found {len(unique_reviews)} new reviews..")

                try:
                    scrollable_element = driver.find_element(By.CSS_SELECTOR, "div.a50Erb.Dy8Cb.k49Klf.d58")
                    driver.execute_script("arguments[0].scrollTop += 2000", scrollable_element)
                    time.sleep(3)
                except Exception as e:
                    print(f"⚠️ Error during scrolling: {e}")

                try:
                    WebDriverWait(driver, 10).until(
                        lambda d: len(extract_reviews()) > len(current_reviews)
                    )
                except Exception:
                    print("⚠️ No new review loaded after scrolling. Retrying...")
                    retry_count += 1
                    if retry_count > 3:
                        break
                    time.sleep(10)

                scroll_count += 1

            reviews = list(all_reviews)

            # Extract review dates and ratings
            try:
                date_elements = driver.find_elements(By.CSS_SELECTOR, "span.raggle")
                rating_elements = driver.find_elements(By.CSS_SELECTOR, "span.ku97ic")
            
```

Objective:

Extract customer reviews of major bank branches across Morocco from Google Maps.

Tools Used:

- Selenium – for automated browser interaction
- BeautifulSoup – for parsing HTML content

Process Steps:

1. Identify Target Locations:

- Gather a list of major Moroccan bank agencies and their branches.
- Use location data and branch names as search queries on Google Maps.

2. Scraping Data:

- Use **Selenium** to navigate dynamically loaded Google Maps review pages.
- Use **BeautifulSoup** to parse and extract the following data:
 - Bank Name
 - Branch Name
 - Geographic Location (latitude/longitude or address)
 - Review Text
 - Star Rating
 - Review Date

3. Data Storage:

- Store the extracted raw data in **JSON** and/or **CSV** format for further transformation.

	Bank name	Branch name	Location	Review text	Rating	Review date
0	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	Catastrophe !\nJ'ai d'abord demandé à ce qu'on...	2,1	2020
1	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	Je suis un client de la BMCE, et très étonné d...	2,1	2019
2	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	Le client doit s adapter avec les horaires de ...	2,1	2024
3	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	Pas mal comme banque	2,1	2021
4	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	ils ne décroche pas le telephone et ne repond ...	2,1	2022
5	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	Je trouve le coin agréable	2,1	2021
6	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	Service zbaaaaala	2,1	2022
7	BANK OF AFRICA, Agence Abdelmoumen, Casablanca.	BANK OF AFRICA, Agence Abdelmoumen, Casablanca...	H99F+3CV, Rue Pasquier, Casablanca 20250	Si je pouvais donner moins d'une étoile, je di...	2,1	2023
8	بنك أفريقيا	بنك H848+H34, Casablanca	H848+H34, Casablanca	Meilleur banque	3,0	2018
9	بنك أفريقيا	بنك H848+H34, Casablanca	H848+H34, Casablanca	Fidélité Casablanca	3,0	2018
0	بنك أفريقيا	بنك H848+H34, Casablanca	H848+H34, Casablanca	Non	3,0	2021
1	بنك أفريقيا	بنك H848+H34, Casablanca	H848+H34, Casablanca	Non	3,0	2024
2	بنك أفريقيا	بنك H848+H34, Casablanca	H848+H34, Casablanca	Non	3,0	2018
3	بنك أفريقيا	بنك H848+H34, Casablanca	H848+H34, Casablanca	Non	3,0	2018
4	بنك أفريقيا	بنك H848+H34, Casablanca	H848+H34, Casablanca	Non	3,0	2018

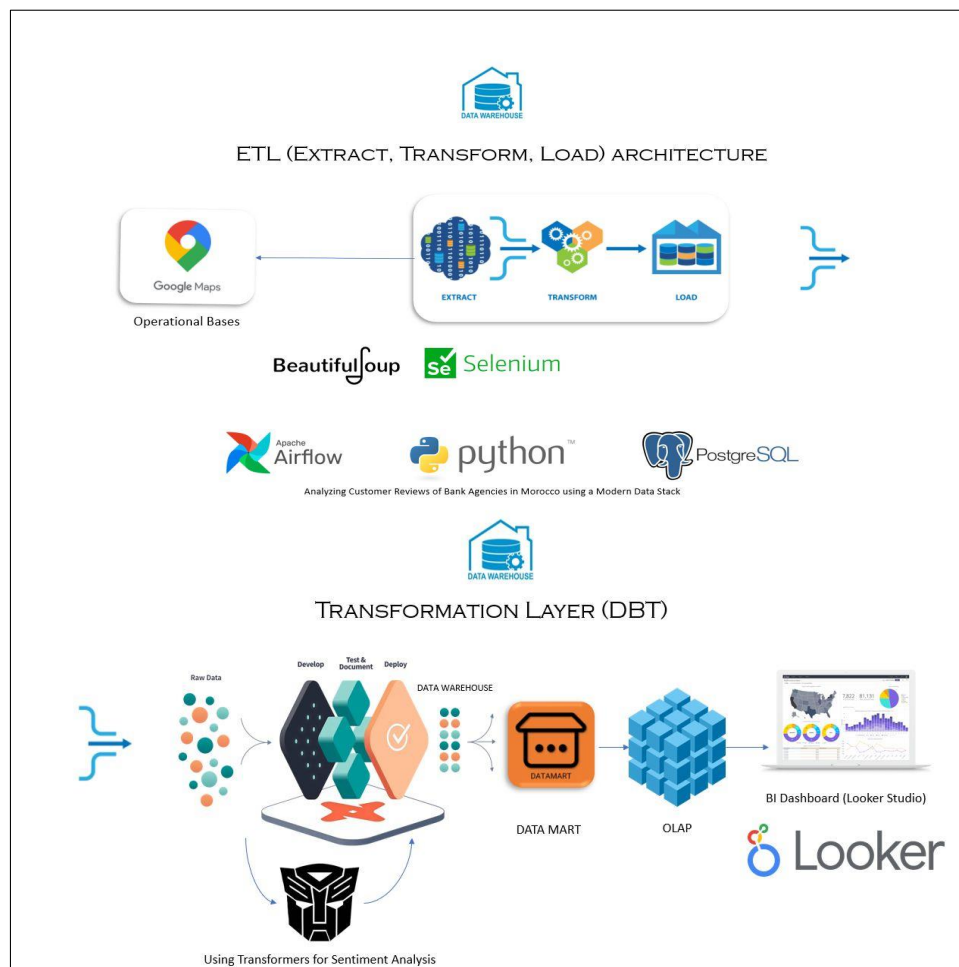
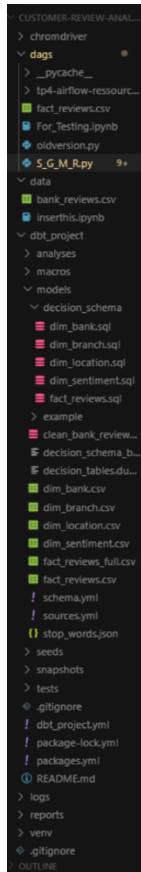


Figure 5: Data Cleaning and Transformation Pipeline illustrating steps like deduplication, text normalization, sentiment analysis, and topic modeling.

Project Structure



The S_G_M_R.py DAG orchestrates a full workflow for data extraction and preprocessing. It contains all the tasks necessary to:

- Extract raw data from various sources (databases, APIs, files, etc.)
- Clean and preprocess the extracted data (handling missing values, filtering, normalization, etc.)
- Prepare the data for downstream tasks such as modeling or analysis

Each task in the DAG represents a specific step in the extraction and preprocessing pipeline, ensuring that the data flows through a well-defined sequence. This modular setup allows for easier debugging, monitoring, and maintenance of the entire data pipeline.

By centralizing these tasks in one DAG file, S_G_M_R.py provides a clear and organized view of the end-to-end data preparation process.



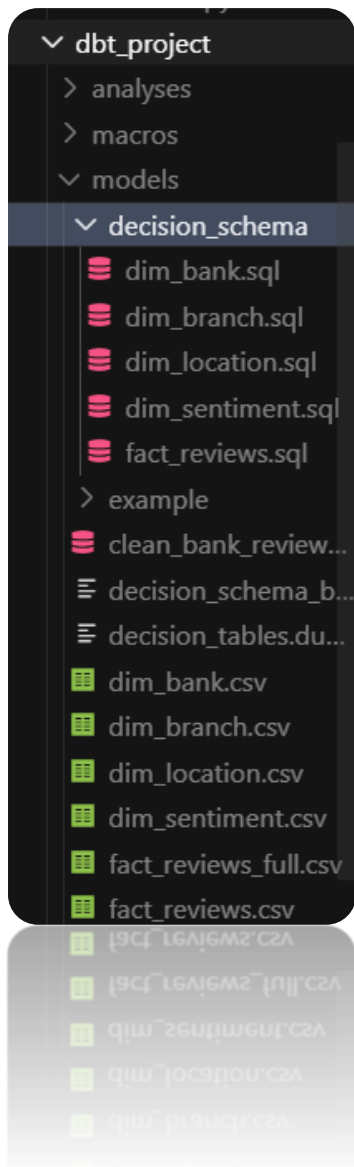
Le projet utilise DBT pour structurer les données selon un modèle en étoile, stocké dans le schéma decision_schema. Ce schéma comprend :

- dim_bank.sql : les informations des banques
- dim_branch.sql : les agences bancaires
- dim_location.sql : les localisations géographiques
- dim_sentiment.sql : les sentiments extraits des avis
- fact_reviews.sql : les avis clients avec les notes, dates et liens vers les dimensions

Les modèles DBT permettent de transformer les données brutes issues du scraping en données analytiques prêtes à l'emploi dans PostgreSQL.

Le chargement et la transformation sont automatisés avec Apache Airflow, qui orchestre les étapes du pipeline : extraction, transformation et mise à jour des tables.

DBT_project



Le projet utilise **DBT** pour structurer les données selon un **modèle en étoile**, stocké dans le schéma `decision_schema`. Ce schéma comprend :

- `dim_bank.sql` : les informations des banques
- `dim_branch.sql` : les agences bancaires
- `dim_location.sql` : les localisations géographiques
- `dim_sentiment.sql` : les sentiments extraits des avis
- `fact_reviews.sql` : les avis clients avec les notes, dates et liens vers les dimensions

Les modèles DBT permettent de transformer les données brutes issues du scraping en données analytiques prêtes à l'emploi dans PostgreSQL.

Le chargement et la transformation sont **automatisés avec Apache Airflow**, qui orchestre les étapes du pipeline : extraction, transformation et mise à jour des tables.

This function `get_sentiment` performs sentiment classification on a customer review by using a pre-trained transformer model through the Hugging Face `pipeline`. The model used here is `"nlptown/bert-base-multilingual-uncased-sentiment"`, which predicts sentiment on a 1 to 5 star scale.

- The function takes a review text and the loaded `sentiment_pipeline` as inputs.
- It runs the review through the pipeline to get the predicted label, which is typically a string like "5 stars", "4 stars", etc.
- Based on the predicted star rating in the label, it maps the result to a simplified sentiment category:
 - "Positive" if the label contains 4 or 5,
 - "Neutral" if it contains 3,
 - "Negative" otherwise (1 or 2 stars).

This approach leverages the transformer's fine-tuned multilingual BERT model to provide robust sentiment analysis and then translates the detailed star ratings into a simpler sentiment classification.

```
# Apply the sentiment analysis
df["sentiment"] = df["review"].apply(lambda x: get_sentiment(x, sentiment_pipeline))
```

To extract the topic words, the code first fits an LDA model on the review data transformed into a document-term matrix. After fitting:

- It accesses `lda.components_`, which contains the word-topic distributions — basically, how strongly each word is associated with each topic.
- For each topic, it sorts the words by their importance (weight) in descending order.
- Then it selects the top 5 words with the highest weights for each topic.
- These top words are mapped back to their actual terms using the vectorizer's vocabulary (`feature_names`).
- Finally, the top words per topic are combined into a comma-separated string (`topic_labels`), which can be assigned back to reviews to indicate the key words representing their assigned topic.

This way, the code obtains meaningful keywords that characterize each topic learned by the LDA model.

Extract common topics using NLP (Latent Dirichlet Allocation - LDA). and language detection

```
try:
    valid_reviews = df["review"].dropna()
    vectorizer = CountVectorizer(stop_words="english")
    X = vectorizer.fit_transform(valid_reviews)

    lda = LatentDirichletAllocation(n_components=5, random_state=42)
    lda.fit(X)

    topics = lda.transform(X).argmax(axis=1)
    df.loc[df["review"].notnull(), "topic_number"] = topics

    feature_names = vectorizer.get_feature_names_out()

    topic_keywords = {
        topic_idx: [feature_names[i] for i in topic.argsort()[: -6: -1]]
        for topic_idx, topic in enumerate(lda.components_)
    }
    topic_labels = {
        idx: ", ".join(words)
        for idx, words in topic_keywords.items()
    }

    df["topic_words"] = df["topic_number"].apply(lambda x: topic_labels.get(int(x), "Unknown"))
    df["topic_meaning"] = df["topic_words"]

except Exception as e:
    print(f"Topic modeling failed: {e}")
    df["topic_number"] = -1
    df["topic_words"] = "Unknown"
    df["topic_meaning"] = "Unknown"

df.to_sql("final_reviews", con=engine, if_exists="replace", index=False)
print("✅ Post-processing complete!")
```

```
bljuf(, b0zf-bl0ccz2ju8 c0wbj0fc;u)
qt'f0'z0j(,tju8j'lel7em2, cou=eu8ju6' fl'cx7z7z='lebj0ce,' fuqcx=79j2c)

qt['_robjc'w0euju8_] = _fukuoou_
qt['_robjc'w0lq3_] = _fukuoou_
qt['_robjc'um0oc1_] = -1
b0z0j(,ld0r0 w0c0j708 l0r70r0 '0l'1
```

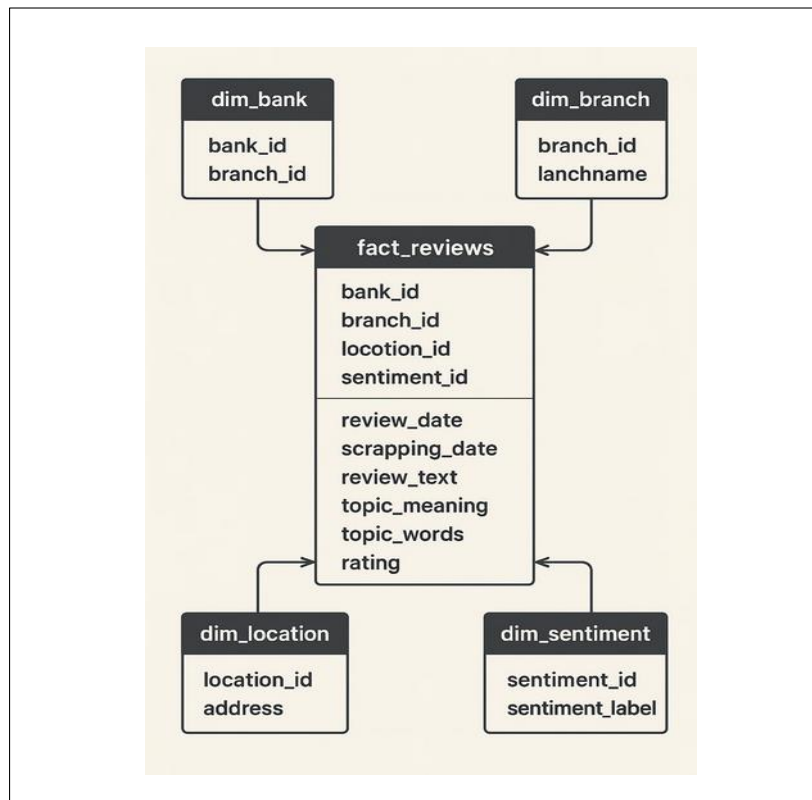


Figure 6: Star Schema for Customer Reviews Analytics, showing the central fact table linked to multiple dimension tables for efficient querying and reporting.

Dim_bank

	bank_id	bank_name
0	1	banque populaire
1	2	Umnia Bank

Dim_location

location_id	address
0	1 9 rue jabal bouiblanc valde ould amir street r...
1	2 bd chefchaoui casablanca 20250
2	3 cited extension 359 amal 4 rabat 10150
3	4 x438jx3 victoria
4	5 avenue dalger rue kairaouane rabat 10020
5	6 2592r5r pl of russia rabat
6	7 x5w2x8m off de france rabat
7	8 ensemble immobilier mahaj riad 23 rabat 10100
8	9 trade imm 1 sis jaouhara subdivision av hassan...
9	10 x4wc78g rabat
10	11 x5w2x8p av ibn sina rabat
11	12 ii and res walid c angle avenue hassan bd ab...
12	13 no 7 av assalam salé
13	14 angle avenue mohamed diouri and rue hamman el ...

Dim_branch

branch_id	branche_name
0	1 ait lamine shopping bd chefchaoui casablanca ...
1	2 ait lamine shopping avenue dalger rue kairaoua...
2	3 ait lamine shopping 9 rue jabal bouiblanc val...
3	4 banque populaire 2592r5r pl de russia rabat
4	5 ait lamine shopping extension cited 359 amal 4...
5	6 banque populaire x5w2x8p off ibn sina rabat
6	7 ait lamine shopping together real estate mahaj...
7	8 banque populaire x4wc78g rabat
8	9 ait lamine shopping n 7 bc salty
9	10 ait lamine shopping ii and res walid c angle...
10	11 banque populaire x438jx3 rabat
11	12 ait lamine shopping angle avenue mohamed diour...
12	13 banque populaire x5w2x8m av de france rabat
13	14 ait lamine shopping commerce imm 1 sis jaouhar...

Dim_sentiment

sentiment_id	sentiment_label
0	1 Negative
1	2 Positive
2	3 Neutral

Fact_reviews

	bank_id	branch_id	location_id	sentiment_id	review_date	review_text	topic_meaning	topic_words	rating
0	2	9	13	2	2024-11-18	umnia bank offre excellent service interface L...	ATM Issues	agence, service, salé, compte, pire	5
1	2	1	2	1	2025-03-18	soit gardez sécurité soit retrancheront reven...	Customer Service	services, sécurité, soit, personnel, banque	1
2	2	9	13	2	2024-12-18	satisfaite experience umnia bank bon service é...	Loan Problems	bon, service, banque, bank, expérience	5
3	2	14	9	2	2025-03-18	bon accueil etbon service	Loan Problems	bon, service, banque, bank, expérience	1
4	2	2	5	2	2024-06-18	employés veulent servir buzz donnera informations	ATM Issues	agence, service, salé, compte, pire	5
...	...	...	...	...	...	...	...	...	...
86	2	7	8	3	2018-03-18	bank umnia hay riad avenue attine	Bank Fees	mr, remercier, tiens, comment, service	4
87	2	2	5	1	2025-03-18	dommage quaucune agences nest ouverte samedi	ATM Issues	agence, service, salé, compte, pire	1
88	1	11	4	1	2025-01-18	numéro téléphone mal rempli	Online Banking Issues	professionnalisme, service, agence, umnia, banque	1
89	1	11	4	1	2022-03-18	lincompétence règle agence professionnalisme ...	Online Banking Issues	professionnalisme, service, agence, umnia, banque	1
90	2	3	1	2	2018-03-18	site web estil sorti	Bank Fees	mr, remercier, tiens, comment, service	5

timent scores. Visualizations include time series, heatmaps, word clouds, and bar charts, allowing stakeholders to identify patterns and anomalies intuitively.

4. **Deployment and Automation:** To ensure sustainability, the entire data pipeline is

- 5. deployed within a production environment managed by Apache Airflow. Sched-uled DAGs execute weekly data extraction, transformation, and loading tasks, guar- anteeing that the dashboards reflect the most current customer feedback on a weekly basis. Error handling and alert mechanisms notify the team of any pipeline disrup- tions, enabling timely remediation.
- 6. **Evaluation and Continuous Improvement:** Post-deployment, the system’s outputs are evaluated with bank stakeholders through feedback sessions. Insights gathered guide iterative improvements in data processing algorithms, dashboard features, and scope expansion such as integration of additional data sources or multilingual support.

This phased approach balances technical complexity with business relevance, ensuring that the project delivers tangible value within a realistic timeframe.

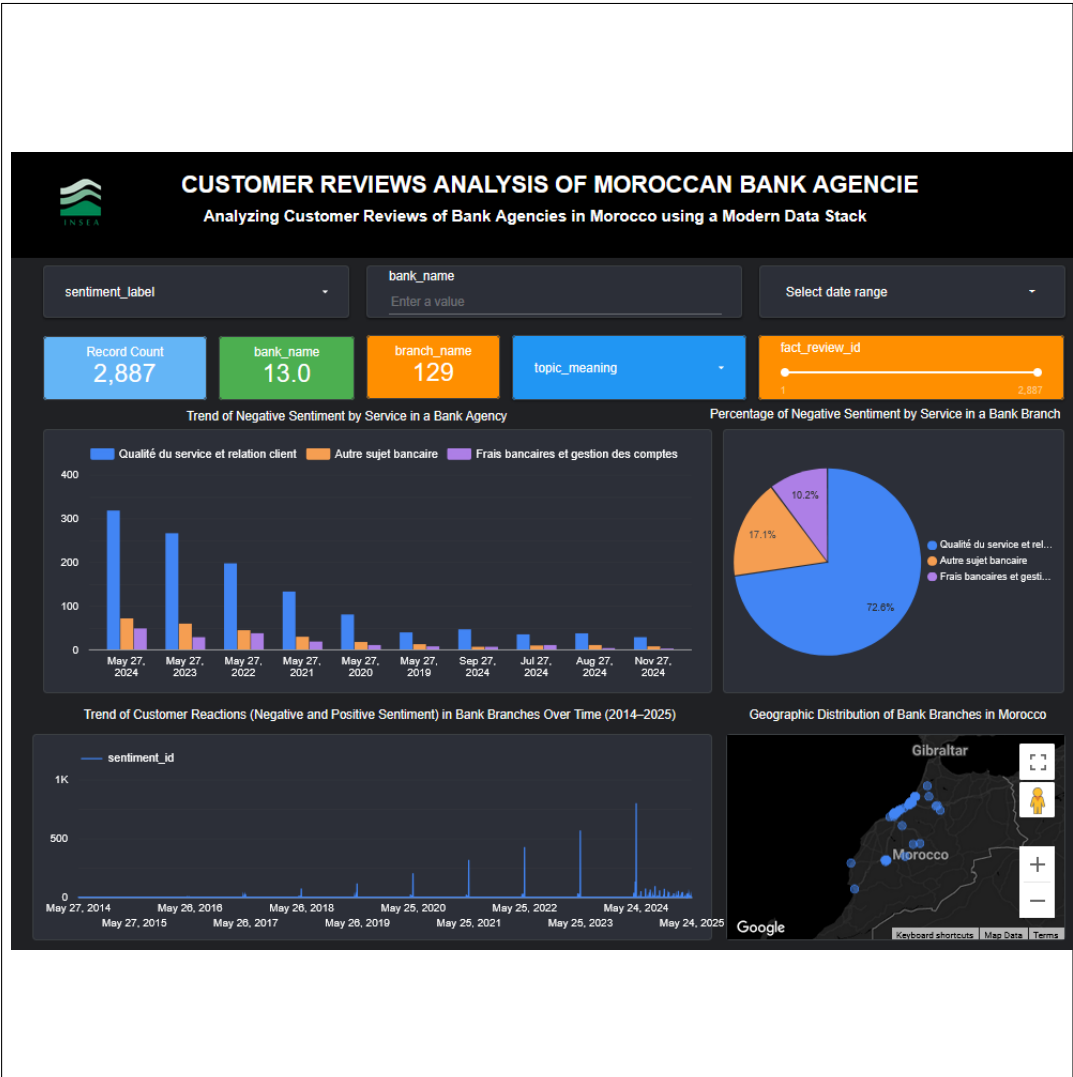


Figure 7: Sample Looker Studio Dashboard illustrating sentiment trends, branch rankings, and topic distributions based on customer reviews.

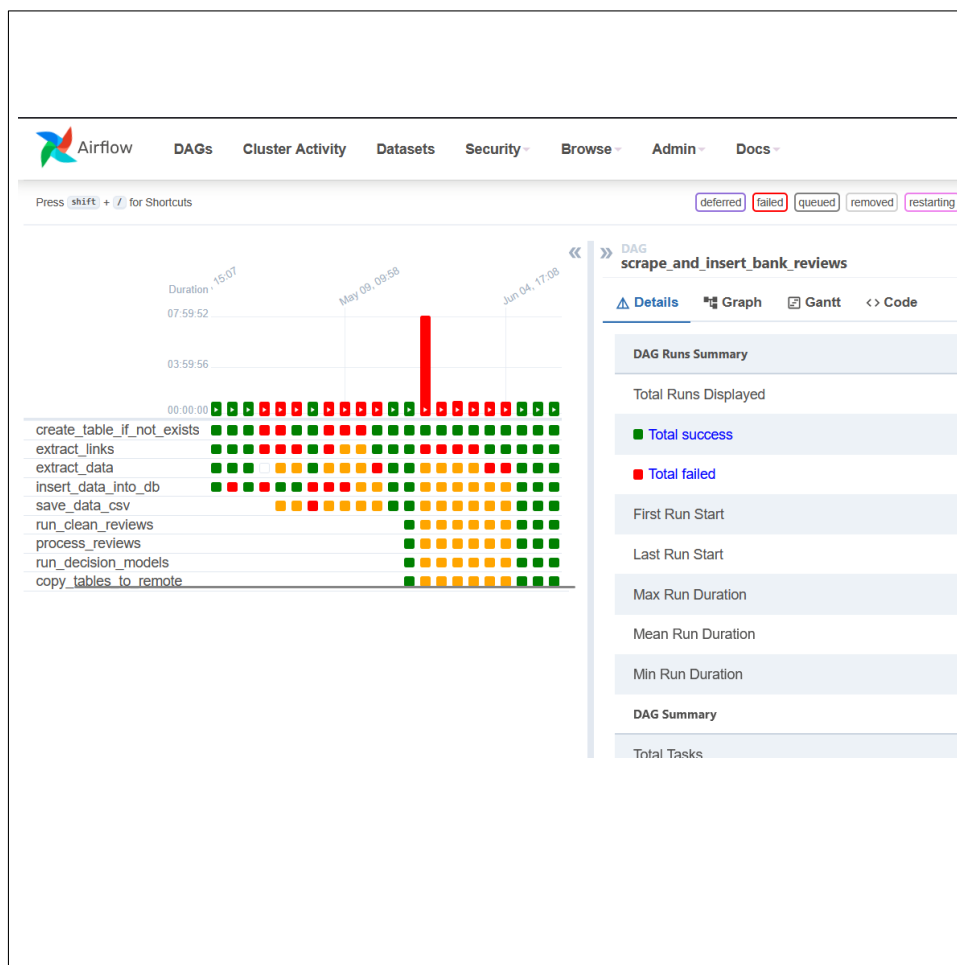


Figure 8: Apache Airflow DAG illustrating the orchestration of ETL tasks on a weekly schedule with monitoring and alerting capabilities.

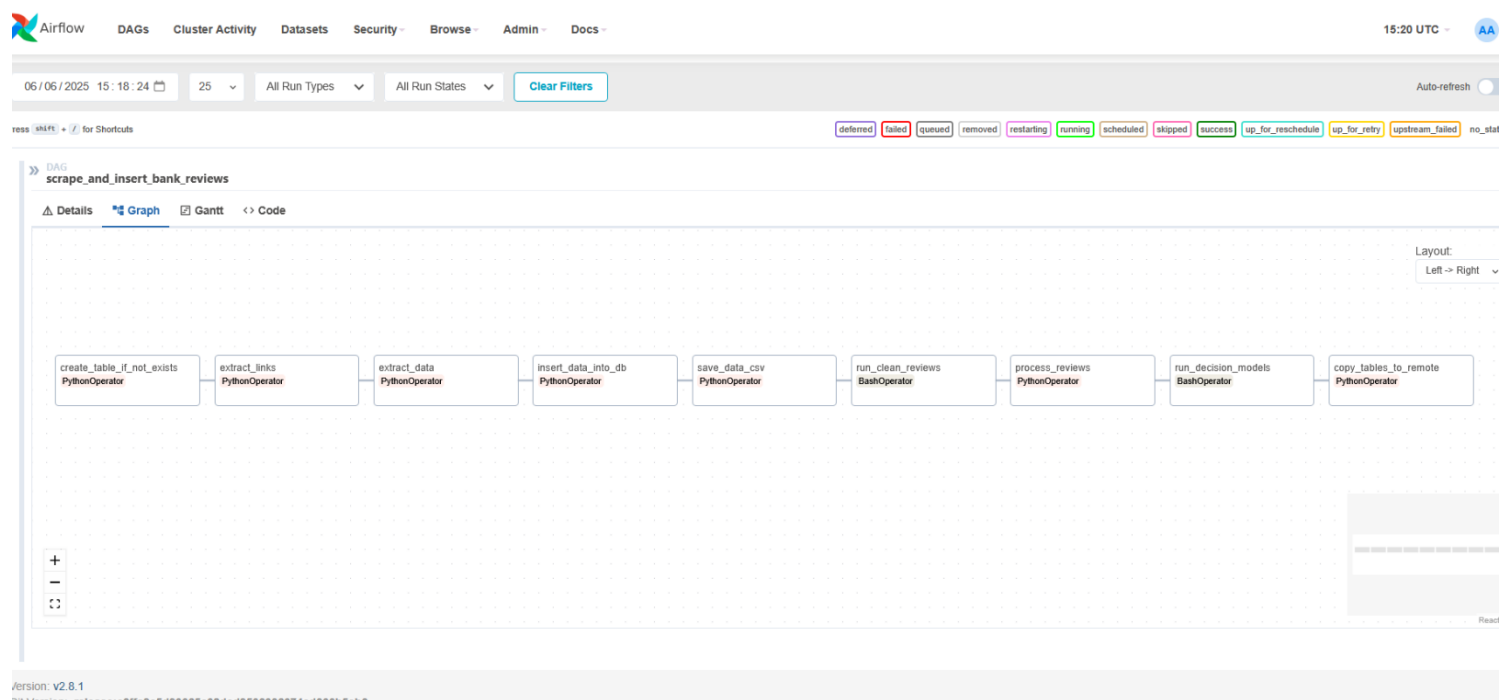


Figure 9: Continuous Improvement Cycle based on stakeholder feedback to refine data processes and system functionalities.

5 Chapter 3 – Data Analysis and Visualization

6 Introduction

The Data Analysis and Visualization phase is a critical component of the project, transforming processed and modeled data into actionable insights. By leveraging interactive dashboards and visual representations, stakeholders can intuitively explore trends, identify pain points, and make data-driven decisions to enhance customer experience across Moroccan bank agencies.

7 Key Analytical Objectives

The analysis focuses on addressing the following questions:

- **Sentiment Distribution:** What is the overall sentiment (positive, neutral, negative) of customer reviews across different banks and branches?
- **Temporal Trends:** How has customer sentiment evolved over time (2014–2025)? Are there seasonal or event-driven fluctuations?
- **Service-Specific Insights:** Which banking services (e.g., customer service quality, fees, digital offerings) receive the most criticism or praise?
- **Branch Performance:** How do branches compare in terms of aggregated sentiment scores and review volumes?
- **Topic Prevalence:** What are the most frequently discussed topics in reviews, and how do they correlate with sentiment?

8 Visualization Components

The Looker Studio dashboards incorporate the following visualizations to address the above objectives:

8.1 Sentiment Overview

- **Pie Charts/Bar Graphs:** Display the proportion of positive, neutral, and negative reviews for selected banks or branches.
- **Filters:** Allow stakeholders to segment data by bank name, branch, or date range (as shown in the mockup: “Select date range”).

8.2 Trend Analysis

- **Time Series Graphs:** Illustrate the trend of negative and positive sentiments over time (e.g., “Trend of Customer Reactions Over Time”). Annotations highlight events like policy changes or branch renovations that may impact sentiment.
- **Comparative Lines:** Overlay sentiment trends for multiple banks or services to identify relative performance.

8.3 Service-Specific Breakdown

- **Stacked Bar Charts:** Show the percentage of negative sentiment by service category (e.g., “Frais bancaires et gestion des comptes”).
- **Heatmaps:** Correlate topics (e.g., “Qualité du service”) with sentiment intensity across branches.

8.4 Branch Performance Metrics

- **Ranking Tables:** List branches by aggregated sentiment scores or review volumes.
- **Geospatial Maps:** Pinpoint branches with the highest/lowest satisfaction using location data (e.g., “Gibraltar”, “Morocco” from the mockup).

8.5 Topic Modeling Outputs

- **Word Clouds:** Highlight frequently occurring terms in positive/negative reviews.
- **LDA Visualization:** Display topic clusters (e.g., “Autre sujet bancaire”) and their prevalence.

9 Tools and Techniques

9.1 Sentiment Analysis

- Pretrained NLP models (e.g., VADER, spaCy) classify reviews into sentiment categories.
- Custom fine-tuning ensures accuracy for banking-specific jargon (e.g., “frais cachés” → negative).

9.2 Topic Modeling

- Unsupervised algorithms (LDA, NMF) extract latent themes from review text.
- Manual labeling refines topics (e.g., merging “long wait times” and “staff responsiveness” into “Qualité du service”).

9.3 Interactive Features

- Dynamic Filtering: Users adjust parameters (e.g., date range, bank name) to drill into specific subsets.
- Tooltips: Hover-over actions reveal metadata (e.g., review excerpts, branch address).

10 Insights from Mockup Data

The provided mockup (Figure 10) reveals preliminary insights:

- Record Count: 2,887 reviews analyzed, spanning 129 branches and 13 banks.
- Negative Sentiment Drivers: Services like “Qualité du service et relation client” dominate negative feedback.
- Temporal Gaps: Dates in the mockup (“May 32, 2020”) suggest placeholder values requiring data validation.

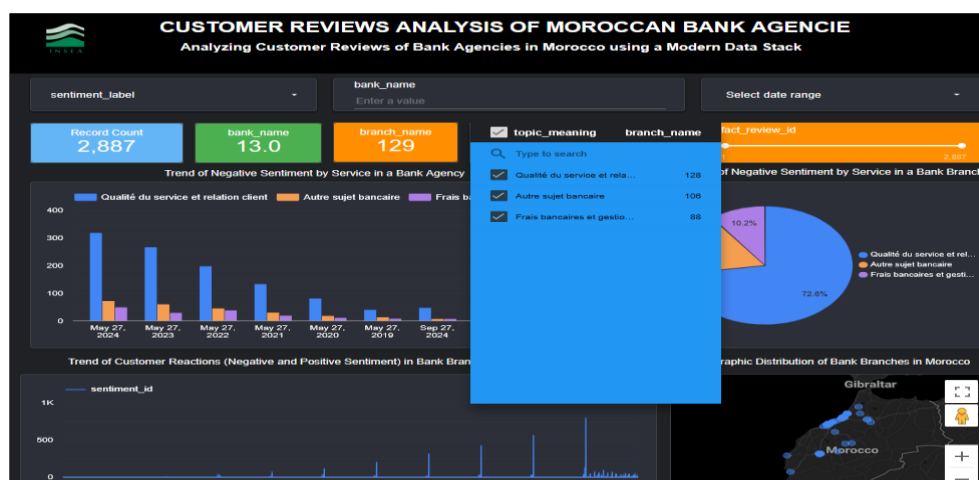


Figure 10: Mockup of Looker Studio dashboard for customer review analysis.

11 Challenges and Solutions

- **Data Noise:** Irrelevant reviews (e.g., spam) are filtered using NLP-based heuristics.
- **Multilingual Text:** Reviews in Arabic/French are translated and standardized.
- **Visual Clutter:** Dashboards prioritize simplicity with collapsible sections and hierarchical filters.

12 Technologies used

TECHNOLOGIES AND TOOLS USED TO BUILD THE PROJECT



1. **Apache Airflow**
An open-source platform to programmatically schedule, orchestrate, and monitor workflows (e.g., data pipelines).
2. **PostgreSQL**
A powerful, open-source relational database system (RDBMS) known for scalability and support for JSON and complex queries.
3. **DBT (Data Build Tool)**
A transformation tool that enables analytics engineers to transform data in warehouses using SQL, with version control and testing.
4. **Python**
A versatile programming language widely used for data extraction, cleaning, analysis, and automation.
5. **BeautifulSoup**
A Python library for web scraping, parsing HTML/XML to extract data from websites.

Conclusion

This project, “**Analyzing Customer Reviews of Bank Agencies in Morocco using a Modern Data Stack,**” successfully demonstrated the value of leveraging unstructured customer feedback from Google Maps to extract actionable insights for banking institutions. By systematically collecting, cleaning, modeling, and analyzing thousands of customer reviews, we transformed scattered textual data into meaningful metrics reflecting customer satisfaction and branch performance.

The implementation of a modern data architecture combining cloud-hosted Post-greSQL, DBT for transformation, and Apache Airflow for orchestration ensured a scalable, maintainable, and automated pipeline. This infrastructure enables weekly updates to keep analyses current and relevant, addressing the dynamic nature of customer sentiment and feedback.

Key achievements include the effective application of advanced Natural Language Processing techniques such as sentiment analysis and topic modeling. These methods uncovered underlying themes in customer opinions—ranging from praise for professional staff to concerns about queue management and digital service quality. The use of a star schema data model facilitated efficient querying and robust reporting, which were visualized through interactive dashboards tailored for banking stakeholders.

Moreover, the deployment of an automated pipeline with error handling and alerting mechanisms guarantees operational reliability and supports continuous improvement through stakeholder feedback loops. This iterative process opens opportunities to expand the system’s capabilities, such as incorporating multilingual support, integrating additional data sources like social media, or enhancing predictive analytics for customer behavior.

In conclusion, this project highlights the immense potential of modern data engineering and machine learning techniques in turning everyday customer interactions into strategic business intelligence. The insights generated empower Moroccan banks to optimize branch operations, improve customer experience, and ultimately strengthen their competitive advantage in an increasingly digital marketplace.

Future Work: Building upon this foundation, future enhancements may include real-time data ingestion for near-instantaneous feedback analysis, deeper sentiment granularity with aspect-based sentiment models, and the integration of chatbot feedback or direct customer surveys to complement Google Maps reviews. Additionally, exploring advanced visualization tools and mobile reporting can further democratize access to these insights across the bank’s organizational levels.

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